# Analysis

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TPOT with linear search space versus ensemble

2000 total evals

Version 1

## Stats Testing

```
## # A tibble: 26 x 9
##    task_id .y.   group1            group2    n1    n2 statistic       p p.signif
##      <int> <chr> <chr>             <chr>  <int> <int>     <dbl>   <dbl> <chr>
##  1    2073 auroc ensemble_random_~ indiv~    21    21       198 2.86e-3 **
##  2  146818 auroc ensemble_random_~ indiv~    21    21       189 9.02e-3 **
##  3  146820 auroc ensemble_random_~ indiv~    21    21       187 1.13e-2 *
##  4  168350 auroc ensemble_random_~ indiv~    21    21       225 1.34e-5 ****
##  5  168757 auroc ensemble_random_~ indiv~    21    21       222 3.15e-5 ****
##  6  168784 auroc ensemble_random_~ indiv~    21    21       228 4.77e-6 ****
##  7  168911 auroc ensemble_random_~ indiv~    21    21       194 4.88e-3 **
##  8  190137 auroc ensemble_random_~ indiv~    21    21       206 8.52e-4 ***
##  9  190146 auroc ensemble_random_~ indiv~    21    21       159 1.37e-1 ns
## 10  190411 auroc ensemble_random_~ indiv~    21    21       228 4.77e-6 ****
## 11  359955 auroc ensemble_random_~ indiv~    21    21       157 1.54e-1 ns
## 12  359956 auroc ensemble_random_~ indiv~    21    21       189 9.02e-3 **
## 13  359958 auroc ensemble_random_~ indiv~    21    21       209 1.23e-3 **
## 14  359959 auroc ensemble_random_~ indiv~    21    21       216 1.31e-4 ***
## 15  359960 auroc ensemble_random_~ indiv~    21    21        11 1   e+0 ns
## 16  359962 auroc ensemble_random_~ indiv~    21    21       209 5.1 e-4 ***
## 17  359963 auroc ensemble_random_~ indiv~    21    21       218 8.39e-5 ****
## 18  359964 auroc ensemble_random_~ indiv~    21    21       210 4.26e-4 ***
## 19  359965 auroc ensemble_random_~ indiv~    21    21       184 1.58e-2 *
## 20  359968 auroc ensemble_random_~ indiv~    21    21       181 2.16e-2 *
## 21  359969 auroc ensemble_random_~ indiv~    21    21       231 9.54e-7 ****
## 22  359970 auroc ensemble_random_~ indiv~    21    21       230 1.91e-6 ****
## 23  359971 auroc ensemble_random_~ indiv~    21    21       212 2.93e-4 ***
## 24  359972 auroc ensemble_random_~ indiv~    21    21       229 2.86e-6 ****
## 25  359974 auroc ensemble_random_~ indiv~    21    21       197 3.28e-3 **
## 26  359975 auroc ensemble_random_~ indiv~    21    21       205 1.98e-3 **
```

## Medians

```
## # A tibble: 26 x 3
##    task_id individual_tpot ensemble_random_100
##      <int>           <dbl>               <dbl>
##  1    2073           0.875               0.890
##  2  146818           0.935               0.935
##  3  146820           0.997               0.997
##  4  168350           0.963               0.966
##  5  168757           0.785               0.795
##  6  168784           0.965               0.968
##  7  168911           0.873               0.874
##  8  190137           0.918               0.930
##  9  190146           0.960               0.964
## 10  190411           0.920               0.922
## 11  359955           0.742               0.748
## 12  359956           0.930               0.934
## 13  359958           0.939               0.942
## 14  359959           0.743               0.754
## 15  359960           1                   1
## 16  359962           0.836               0.846
## 17  359963           0.995               0.996
## 18  359964           0.993               0.994
## 19  359965           1.00                1.00
```

```
## 20  359968           0.919              0.929
## 21  359969           0.847              0.851
## 22  359970           0.910              0.921
## 23  359971           0.997              0.997
## 24  359972           0.990              0.993
## 25  359974           0.830              0.873
## 26  359975           0.993              0.996
```